a Construct the measured response

Raw counts

310,385 cells · 8,563 genes
48 gem groups

UMI normalization

$$\widetilde{X}_{ig} = 10^4 X_{ig}^{\mathrm{raw}} / \mathrm{UMI}_i$$

Independent of controls

Pseudobulk + control

$$Y_{bpg} = P_{bpg} - C_{bg}$$

Shared control is explicit

b Four-term comparison

	g_1	g_2	
p_1	+1	-1	$a^\top X_E = 0$
p_2	-1	+1	$h = 0$

c Query eligibility

Query	Identified	Source-safe
Q1 atomic	No	No
Q2 perturbation	No	Yes
Q3 gene	No	No
Q4 four-term	Yes	Yes

100,000 accepted Q4 queries; no extra source rejections